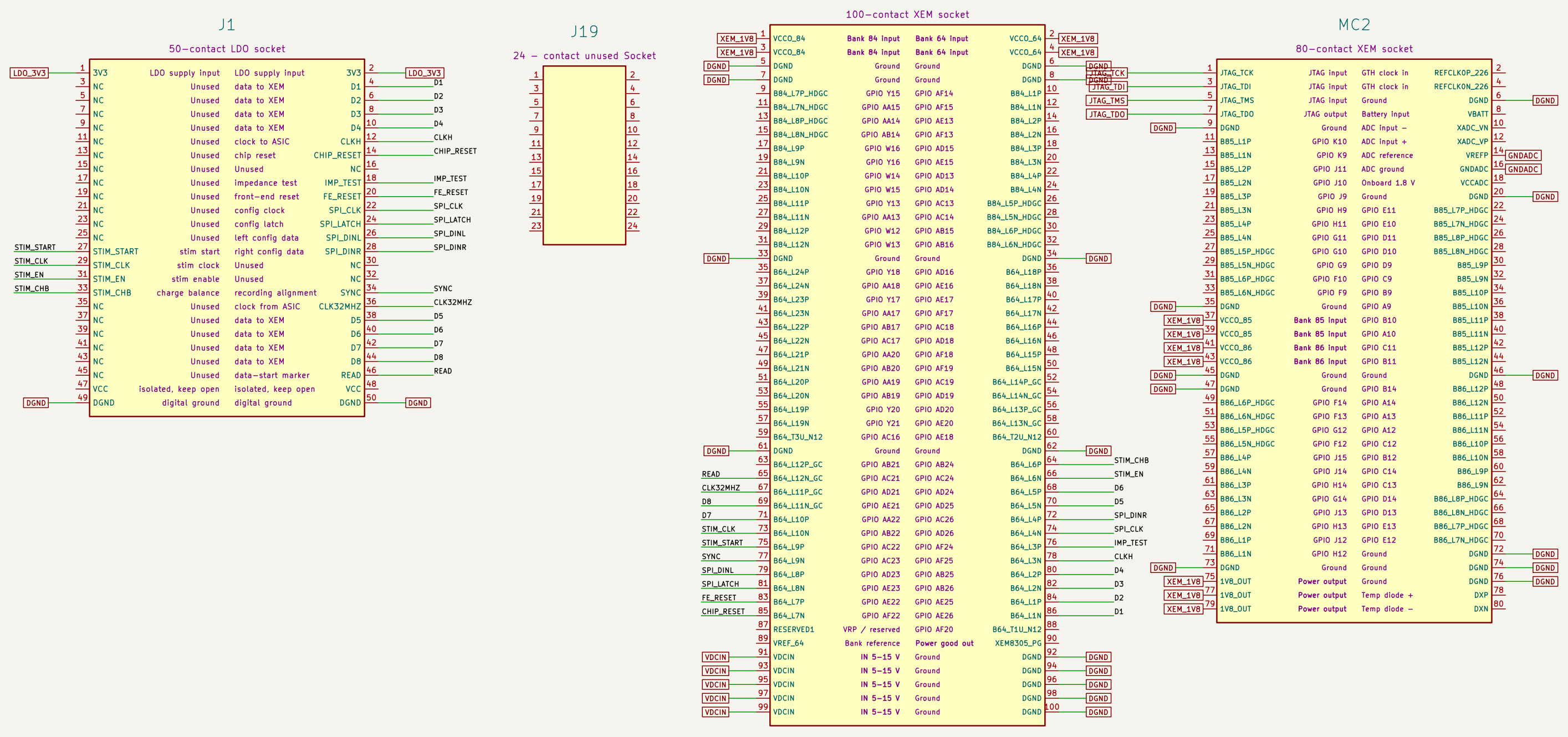
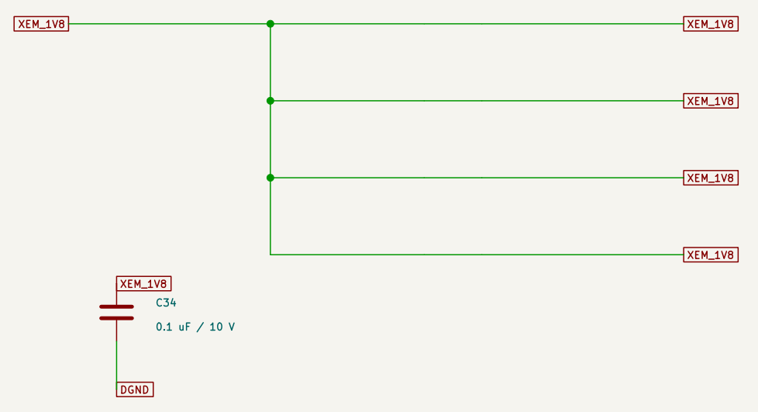
A LDO AND XEM CONNECTORS (J1 / J19 / MC1–MC3)



1. Function: Connect the inherited LDO–board digital interface and 3.3 V feed to the XEM8305; J19 is the retained auxiliary socket.

2. Source / changes: Inherited Gerald LDO pin map; Codex designed this carrier integration, not a manufacturer application circuit. Oct 2 Gerald native source and the V19 source ZIP differ; confirm the physical LDO revision.
Opal Kelly XEM8305 Expansion Connectors, MC1/MC2/MC3 tables: https://docs.opalkelly.com/xem8305/expansion–connectors/
Hirose DF40 Series catalog, Jan 2025, pp5–6,9: https://www.hirose.com/en/product/series/DF40 .
Samtec ERF6 recommended PCB layout, Rev C (18 Feb 2022), sheet1 Fig1 and Table1:
https://suddendocs.samtec.com/prints/erf6–xx–xx.x–x–dv–a–xx–k–xr–footprint.pdf
V26: Codex corrected all six MC1/MC2/MC3 –A locating holes from 1.20 to 0.95 mm; finished NPTH tolerance +/–0.05 mm.
Contact centers, copper pads, circuit pin maps and tracks are retained. Actual assembly fit is not yet verified.
ASIC 1.8 V limits, LDO contact orientation, current budget, mating height and component clearance remain open.

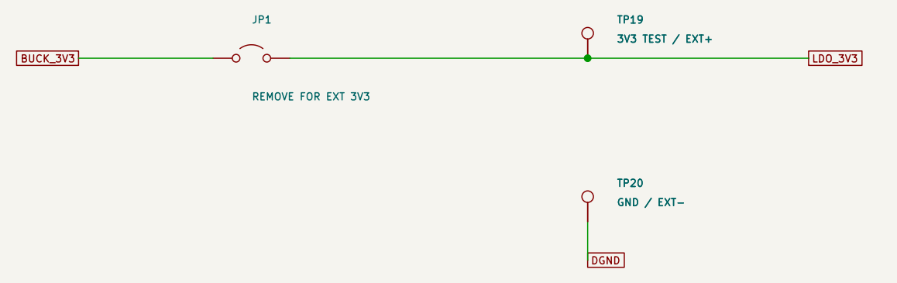
B FPGA BANK POWER (C34 / XEM_1V8)



1. Function: Feed FPGA banks 64/84/85/86 from the module–exported 1.8 V at MC2.75/77/79; C34 provides common 100 nF bypass.

2. Source / changes: Adapted manufacturer BRK8305 A03, main sheet1 (PDFp1), VCCO links and C34:
https://pins.opalkelly.com/downloads/583/download?category_id=3 . Codex integration replaces source R19/R34/R35/R36 (each 0 ohm) with direct wires; C34=0.1 uF/10 V is retained. No added regulator. Module sequencing/current and ASIC–level compatibility are unverified. Module power guidance:
https://docs.opalkelly.com/xem8305/powering–the–xem8305/ .

C EXTERNAL 3.3 V ACCESS (JP1 / TP19 / TP20)



1. Function: JP1 joins BUCK_3V3 to LDO_3V3; removing its shunt lets a regulated external 3.3 V source feed TP19 relative to TP20.

2. Source / changes: Codex test–access circuit, inherited from V18; JP1 now Harwin M20–9990245 with M7566–05 shunt.
Header DRG–00479, issue 23, sheet 2; shunt DRG–00743, issue 13, sheet 2. Dimensions, ratings and mounting pattern:
https://www.harwin.com/products/M20–9990245 ; https://www.harwin.com/products/M7566–05
Harwin recommends 1.02 +/–0.05 mm finished holes. V40 deliberately uses 1.10 mm holes and retained 1.70 mm lands for standard JLC +0.13/–0.08 mm hole tolerance and manual soldering. This is a Codex adaptation of the recommended pattern.
Pad centers, pitch and nets are retained; two adjacent vias move +0.30 mm X and four straight power spurs extend.
Underlying converter reference: BRK8305 A03, sheet 1 (PDF p1); adding JP1/test access before LDO J1.1/.2 is Codex design.
https://pins.opalkelly.com/downloads/583/download?category_id=3
Remove the shunt before external drive. Selected LDO A has CAD body clearance; physical fit, removal access, solder joints and load limits remain unqualified. Header/shunt ratings are 3 A at 25 C, not a validated system–current budget.

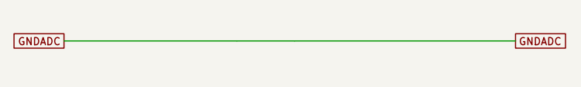
D XEM MOUNTS (MH1–MH4)

MH1
MH2
MH3
MH4
M3 isolated
M3 isolated
M3 isolated
M3 isolated

1. Function: Provide four 3.2 mm plated mounting holes with 6.4 mm isolated copper rings for the XEM stack.

2. Source / changes: Codex mechanical integration, inherited from carrier V18; this is not a datasheet circuit.
Module mounting positions/stack reference: Opal Kelly XEM8305 Specifications, mechanical drawings:
https://docs.opalkelly.com/xem8305/specifications/ . Nominal module connector stack is 5.0 mm; actual standoffs, mating and assembled clearance remain unverified. No electrical components changed in V20.

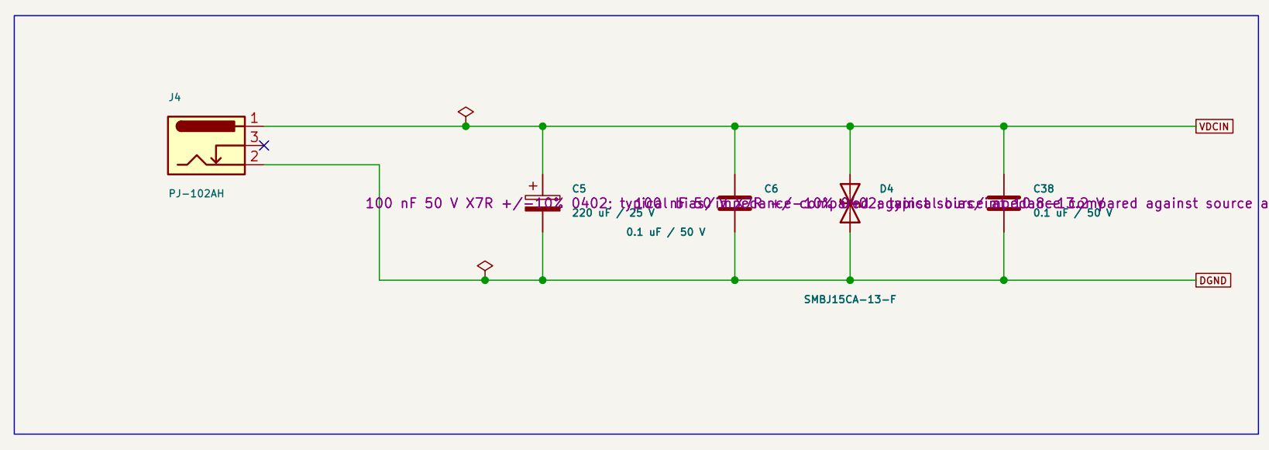
E INTERNAL MONITOR REFERENCE



1. Function: Connect module VREFP to module GNDADC to select the internal FPGA monitor reference; keep GNDADC separate from DGND.

2. Source / changes: Adapted manufacturer BRK8305 A03 main sheet1 (PDFp1), R6 reference link:
https://pins.opalkelly.com/downloads/583/download?category_id=3 . Codex replaces source R6=0 ohm with a direct wire; no extra reference IC or RC network. Opal Kelly Expansion Connectors, MC2.14/.16:
https://docs.opalkelly.com/xem8305/expansion–connectors/ . Module reference behavior is inherited; powered monitor operation remains unverified.

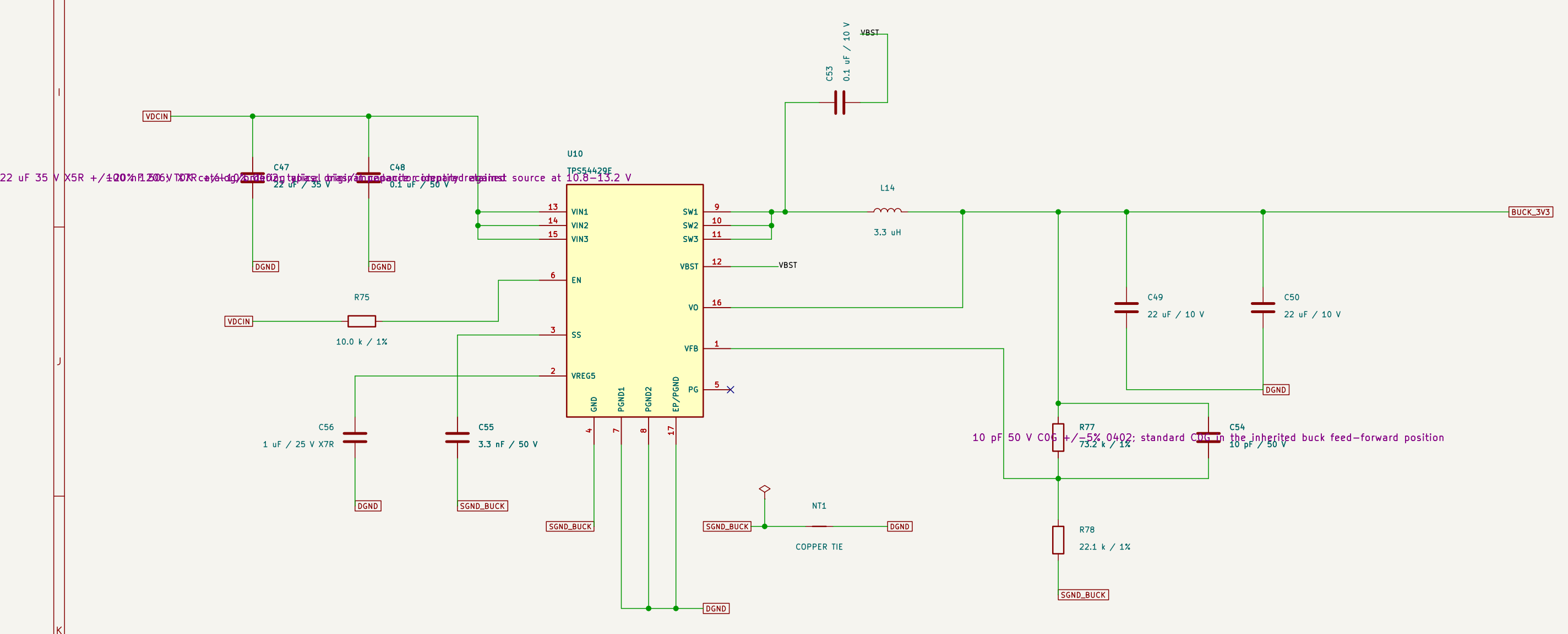
F MAIN 12 V INPUT (J4 / C5 / C6 / C38 / D4)



1. Function: Accept center–positive 12 V, distribute VDCIN to XEM MC1.91/93/95/97/99 and the buck input; C5/C6/C38 bypass the rail and D4 shunts transients.

2. Source / changes: Manufacturer BRK8305 A03 main sheet1 (PDFp1), Power Jack and MC1 bypass:
https://pins.opalkelly.com/downloads/583/download?category_id=3 . Inherited source structure/values: C5=220 uF/25 V, Originally C6=C38=0.1 uF/25 V; D4=SMBJ15CA, J4=PJ–102AH; switched jack pin3 stays open. Codex selects 12 V and omits the alternative J5 input/indicator circuits. TVS clamping, source ripple/load and assembled input protection remain unverified. Module power guidance: https://docs.opalkelly.com/xem8305/powering–the–xem8305/ .
V41: C6/C38 change Taiyo TMK105BJ104KV–F to Samsung CL05B104KB54PNC; 100nF/10% and 0402 retained.
Rating changes 25V X5R to 50V X7R. Samsung product sheet Size/Characteristics and typical bias/impedance data:
https://product.samsungsem.com/mlcc/CL05B104KB54PN.do (accessed 18Oct2026). At 10.8–13.2V, its typical biased capacitance is slightly higher than the original; this is a sourcing comparison, not powered or worst–case qualification.

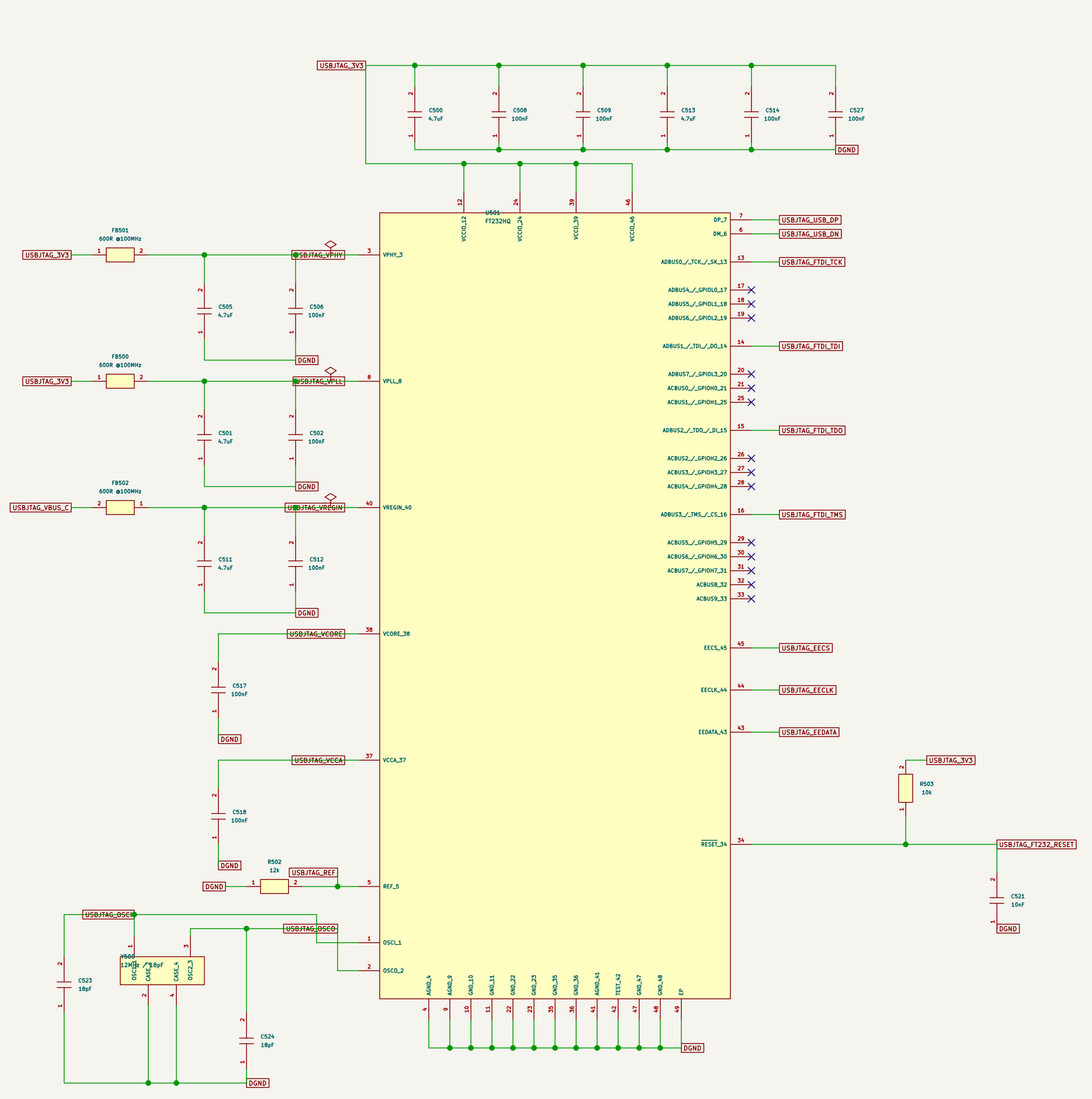
G LDO 3.3 V AND STIMULATION POWER BOUNDARY



1. Function: Convert VDCIN to nominal 3.3 V for the selected LDO A assembly. C47/C48 bypass VIN; L14/C49/C50 filter output; R77/R78/C54 set feedback; C53/C55/C56 provide bootstrap/soft–start/bias. SGND_BUCK joins R78/C55/U10.4 to DGND only at copper NT1. No separate stimulation converters are provided.

2. Source / changes: Adapted BRK8305 A03 main sheet1 (PDFp1), U10 converter: https://pins.opalkelly.com/downloads/583/download?category_id=3 . Inherited Codex adaptation fits R75=10k (source DNP), omits R76=0 ohm, R1/S1 control and PG pullup R79. R77=73.2k, R78=22.1k, C54=10pF, C55=3.3nF, C53=100nF, C47=22uF, C48=100nF, C49=C50=22uF and L14=3.3uH remain. V33 Codex creates local quiet net SGND_BUCK and copper–only NT1 (not a fitted resistor); the ERC flag asserts its DGND feed and adds no hardware. TI TPS54429E SLVSAS0 (Feb2011), p5 RSA pin map and p16 items4/8/11: https://www.ti.com/lit/gpn/TPS54429E .
This is a Codex local implementation of TI single–point SGND/PGND guidance, not a copied RSA layout. C56 changes from LMK105BJ105KV–F 1uF/10V 0402 to Taiyo MSAST168SB7105KTNA01 1uF/25V X7R 0603. It adapts TI EVM SLVU444A (Aug2021), p18 Table7–1 C6 (1uF/16V X7R 0603) to a higher–voltage exact part:
https://www.ti.com/lit/ug/slvu444a/slvu444a.pdf ; https://ds.yuden.co.jp/TYCOMPAS/ut/detail?pn=MSAST168SB7105KTNA01&u=M . Original Source_MPN is retained; this is a replacement, not a renamed part. L14 differs from TI pp13–14/Table1 3.3V row (2.2uH). Loop/startup/ripple/thermal performance remains unverified.
Howard selected LDO A, Downloads folder ending " 2", on 17Oct2026. Its separate J15 INPUT_VDD10, J8 INPUT_VSS10 and AGND connections remain system requirements; 3.3V–only stimulation is unverified.
V41: C47 uses TDK C3216X5R1V226MTJ00E, the same 22uF/35V X5R/1206 under its listed TDK ordering number.
TDK Catalog PN / TDK PN table, letter 12Jan2023, PDF p12: https://media.futureelectronics.com/PCN/95306_SPCN.PDF .
C48 changes TMK105BJ104KV–F to Samsung CL05B104KB54PNC: same 100nF/10%/0402, 25V X5R to 50V X7R.
Samsung product sheet Size/Characteristics and typical curves: https://product.samsungsem.com/mlcc/CL05B104KB54PN.do .
C54 changes KEMET CBR04C100J5GAC to Samsung CL05C100JB5NNNC; same 10pF/5%/50V/C0G/0402, standard series.
Samsung Size/Characteristics: https://product.samsungsem.com/mlcc/CL05C100JB5NNN.do (accessed 18Oct2026).
U10, L14, all values, pad/net assignments and copper are retained. Typical–data substitution screens do not qualify the power loop.

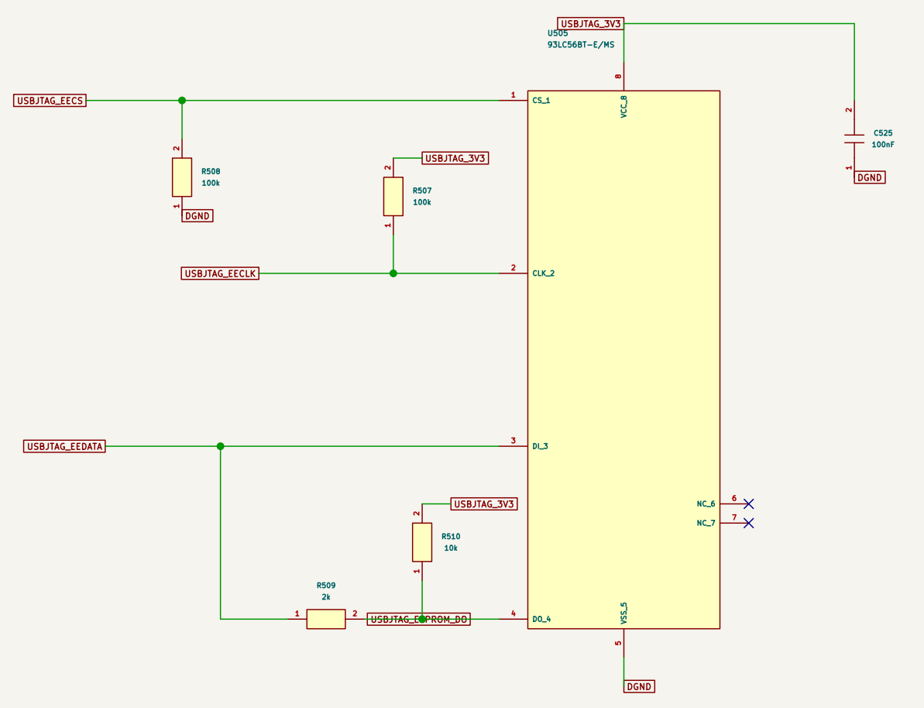
I FT232H BRIDGE AND LOCAL SUPPORT (U501)



1. Function: Turn USB into JTAG using the FT232H MPSSE; local bypass/filter branches power the IC, Y500/C523/C524 provide 12 MHz clock, and R503/C521 support reset.

2. Source / changes: Inherited manufacturer Opal Kelly BRK8305 A03 JTAG sheet8 (PDFp4):
https://pins.opalkelly.com/downloads/583/download?category_id=3 . U501 and all local support R/C values/connections retained, including R502=12k, R503=10k, C521=10nF and C523=C524=18pF. Codex redraws the support wires locally in V18; no electrical change in V20. FTDI FT232H datasheet FT_000288 v2.2, Fig6.1/6.4 pp46/48:
https://ftdichip.com/wp–content/uploads/2020/07/DS_FT232H.pdf . USB/MPSSE operation unverified.
V41: C521 changes KEMET C0402C103K4RACTU to Samsung CL05B103KB5NNNC; same 10nF/10% X7R/0402.
Voltage rating changes 16V to 50V; reset connection and nominal RC timing retained. Samsung Size/Characteristics:
https://product.samsungsem.com/mlcc/CL05B103KB5NNN.do (accessed 18Oct2026). FB500–502 remain exact originals.

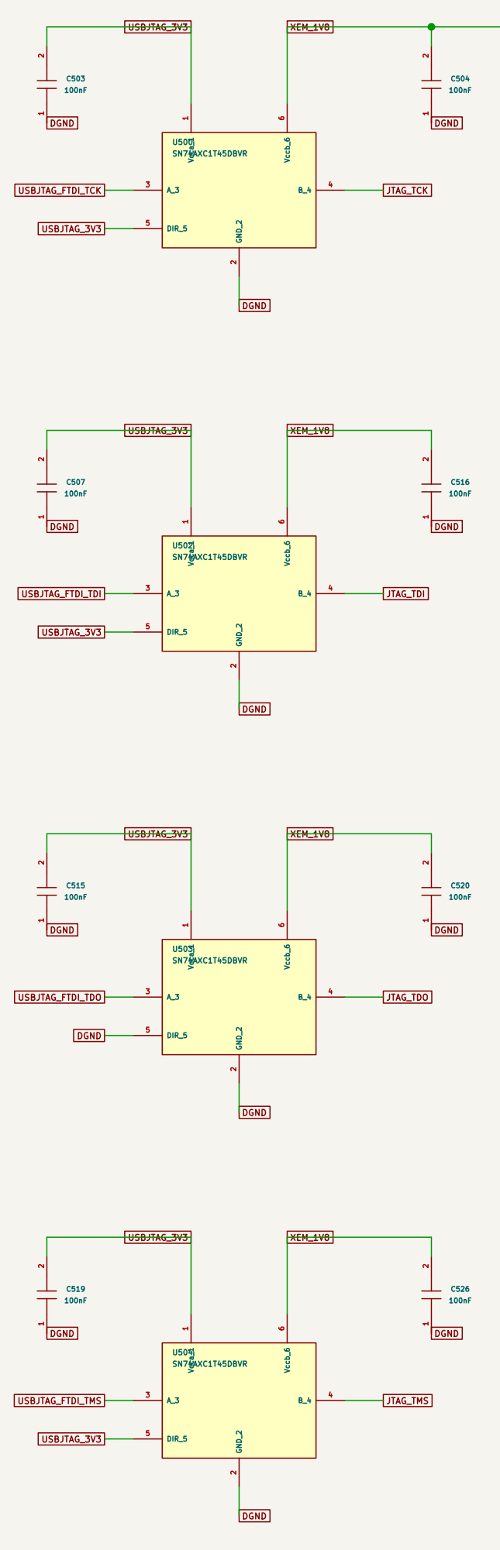
J EEPROM AND LOCAL SUPPORT (U505)



1. Function: Store FT232H configuration in external 16–bit EEPROM; R507/R508 bias clock/select, R509/R510 serve the data path, C525 bypasses 3.3 V.

2. Source / changes: Adapted manufacturer BRK8305 A03 JTAG sheet8 (PDFp4):
https://pins.opalkelly.com/downloads/583/download?category_id=3 . R507=100k, R508=100k, R509=2k, R510=10k and C525=100nF retained. V41 uses blank 93LC56BT–E/MS; the T suffix changes packing only. Source 93LC56B–E/MS–CFG001 was configured; no image was supplied. FTDI FT232H FT_000288 v2.2, section7 p49:
https://ftdichip.com/wp–content/uploads/2020/07/DS_FT232H.pdf . EEPROM programming/unique serial and Vivado discovery remain unverified.
Microchip 93XX56 DS20001794J (2023), pp1–2,20–22,39: LC/B/E/MS ratings, pins, body and T packing code.
https://ww1.microchip.com/downloads/aemDocuments/documents/MPD/ProductDocuments/DataSheets/93AA56X–93LC56X–93C56X–2–Kbit–Microwire–Compatible–Serial–EEPROM–Data–Sheet.pdf
Only packing changes; same LC voltage, B 128x16, E temperature, MSOP8 and blank–device provisioning.

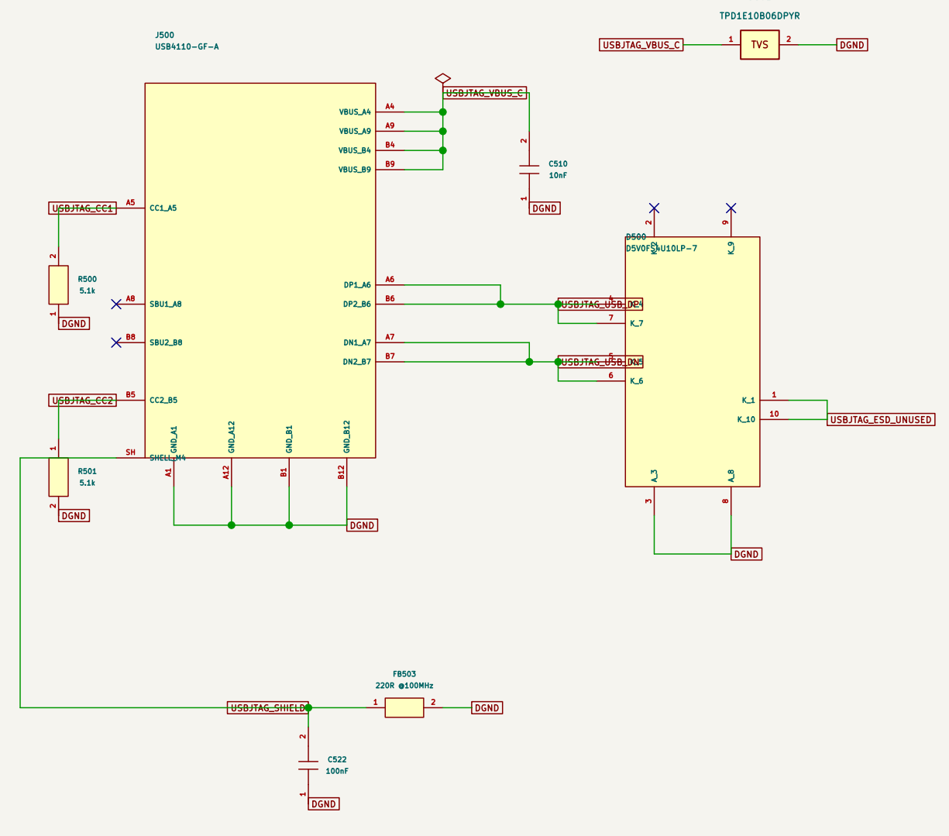
K FOUR JTAG LEVEL SHIFTERS



1. Function: Translate FTDI 3.3 V JTAG to XEM 1.8 V: TCK/TDI/TMS flow A–to–B, TDO B–to–A. Each supply has local 100 nF bypass.

2. Source / changes: Inherited manufacturer BRK8305 A03 JTAG sheet8 (PDFp4), Level Shifters:
https://pins.opalkelly.com/downloads/583/download?category_id=3 . U500/U502/U504 DIR high, U503 DIR low; C503/C504/C507/C516/C515/C520/C519/C526 and extra C528 each 100 nF retained without value/connection changes. TI SN74AXC1T45, SCES882E (Dec2023), Table7–1 p18/Fig8–1 p19: https://www.ti.com/lit/ds/symlink/sn74axc1t45.pdf . Codex adapts net names/contact endpoints and local drawing; powered JTAG and power–order behavior remain unverified.

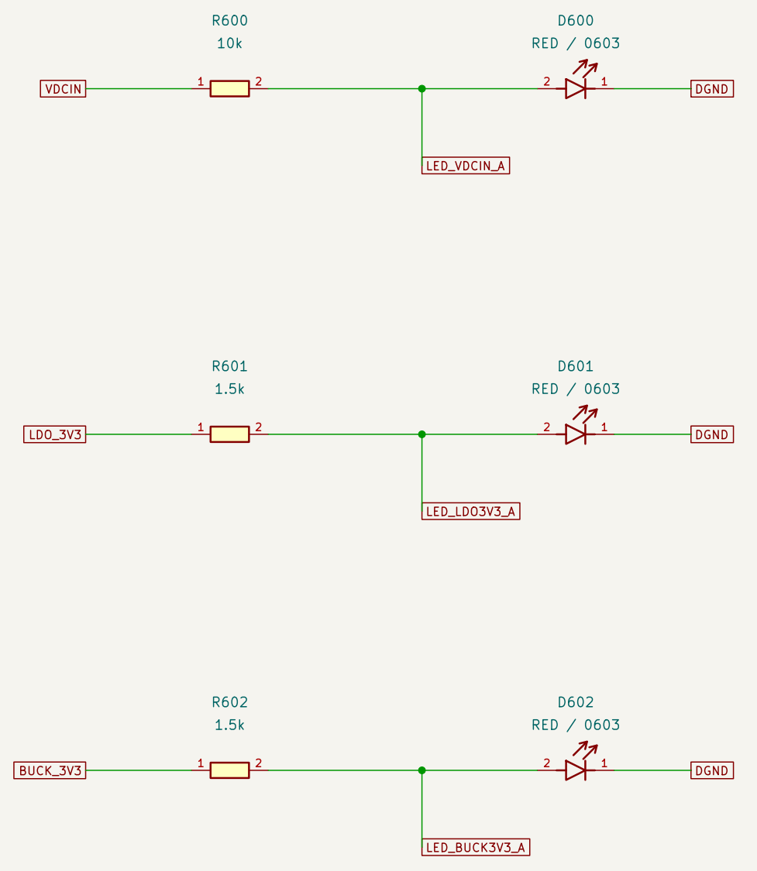
L USB–C INPUT / SEPARATE VBUS TVS / SHIELD (J500 / D500 / D501)



1. Function: Receive USB VBUS and D+/D–; R500/R501 identify a sink. D500 protects D+/D–; D501 is a separate VBUS TVS. C510 bypasses VBUS; C522/FB503 return the shield to DGND.

2. Source / changes: Inherited BRK8305 A03 JTAG sheet 8 (PDF p4), adapted by Codex:
https://pins.opalkelly.com/downloads/583/download?category_id=3
J500 changed from USB4120–03–C to horizontal USB4110–GF–A. R500=R501=5.1k, C510=10nF, C522=100nF and FB503=220 ohm @100MHz retained. Source DNP R504/R505/R506 (0 ohm shield bypasses) and J501 omitted; shield remains on its own net.
V39: D500 pins 2/9 disconnected from VBUS and explicitly NC; D+/D– protection unchanged.
Codex added D501 TPD1E10B06DPYR from VBUS to DGND using TI application guidance.
This is a carrier adaptation, not an unchanged manufacturer reference circuit.
Diodes D5V0FS4U10LP, DS39520 Rev1–2 (Sep2017), pp1–2: pin map / holding ratings.
https://www.diodes.com/datasheet/download/D5V0FS4U10LP.pdf
TI TPD1E10B06, SLLSEB1G (Aug2024), pp3–5,8,11: pins, ratings, mode and layout;
DPY0002A drawing 4224561/C (Jul2024): 0.3x0.5mm lands, 0.7mm pitch.
https://www.ti.com/lit/ds/symlink/tpd1e10b06.pdf
D501 removes the VBUS connection to D500's low–holding channel; this is not ESD qualification.
TVS clamp can exceed C511 6.3V and U501 VREGIN 5.5V operating limits; FB502 attenuation and VREGIN transient withstand need manufacturer guidance and bench validation.
USB enumeration/EMC remain unverified.
V41: C510 changes KEMET C0402C103K4RACTU to Samsung CL05B103KB5NNNC.
Same 10nF/10% X7R/0402; rating 16V to 50V. Samsung product sheet, Size/Characteristics:
https://product.samsungsem.com/mlcc/CL05B103KB5NNN.do (accessed 18Oct2026).

M V25 LOW–CURRENT RAIL CHECK LEDs



1. Function: Passive red indicators show VDCIN, BUCK_3V3 before JP1, and LDO_3V3 after JP1. This distinguishes converter output from the jumper–fed LDO rail. A lit LED does not confirm rail accuracy, power–good, sequencing, or communication. Each branch adds about 1 mA at nominal voltage. No GPIO, USB, 1.8 V rail, or signal is loaded.

2. Source / changes: These series resistor/LED circuits were designed by Codex using component datasheets; they are not a manufacturer application circuit. Newly added to carrier V24: Inherited R600=10k, R601=R602=1.5k (Vishay, 1%): D600–D602=Kingbright APT1608LSECK/J3–PRV. Each rail feeds resistor pin 1; resistor pin 2 feeds LED anode 2; cathode 1 returns to DGND. Existing circuit connections are retained.

Kingbright DSAN8387 / 1203014455, Rev V.6A (25 Mar 2021), pp1–3: ratings, polarity, and recommended solder pattern.
https://www.kingbrightusa.com/images/catalog/SPEC/APT1608LSECK–J3–PRV.pdf
Vishay D/CRCW e3, document 20035, Rev 14–Apr–2026, pp1–3: ratings, standard 0805 operation, and ordering code.
https://www.vishay.com/docs/20035/dcrcwe3.pdf

Nominal currents are calculated, not measured. LED brightness at low current, actual rail extremes, assembly polarity, and powered rail/ripple/thermal behavior remain unverified.
Remove JP1 before externally driving LDO_3V3.
V41: R601/R602 now UNI–ROYAL 0805W8F1501T5E, 1.5k/1%, 100ppm/K, 125mW standard, 0805. At 3.63V and shorted LED, calculated dissipation is 8.87mW.
Royalohm Thick Film Chip Resistors, PDF pp1–3 (printed pp10–12): ratings, dimensions, order code.
https://www.royalohm.com/assets/pdf/products/smd/1.pdf
General–purpose grade; no AEC–Q200 claim.
Connections, resistor value and copper remain unchanged.

Approved sourcing metadata only; all V40 copper, pads, placement and saved fills retained

Sheet: /
File: CARRIER_XEM8305_V41_Sourcing_Alternates.kicad_sch

Title: XEM8305 Carrier – V41 Sourcing Equivalents; V40 Routing

Size: User | Date: 2026–10–18 | **Rev: V41**
KiCad E.D.A. 10.0.6 | Id: 1/1